OverTheWire: Bandit Level 0–10 Walkthrough

share my experience solving The Bandit wargame from OverTheWire. For a brief overview, the OverTheWire community offers hands-on games designed to help us learn and practice security concepts.

To solve this wargame, I used SSH from my Windows terminal.

Level 0
Level 0 is the introduction level and the goal is to log into the game using SSH.

Here is the provided SSH credential:

Host: bandit.labs.overthewire.org
Port: 2220
Username: bandit0
Password: bandit0

I simply run this SSH command to log into the SSH, then fill the password.

ssh bandit0@bandit.labs.overthewire.org -p 2220

Level 0 → 1
We need to get into the next level by logging into the same SSH using username bandit1. However, the password for that username is located inside the readme file in the home directory.

Because the starting point of the SSH is in the home directory (~/), you can directly use cat command to read the readme file.

cat readme

After getting the password, you can continue to the next level. Don’t forget to store the achieved password in the safe place, thus you don’t need to restart the challenge from zero if you want to continue later.

ssh bandit1@bandit.labs.overthewire.org -p 2220

Level 1 → 2
The password is stored inside a file called - in the home directory.

Because the file is stored in a dashed filename, I used the cat ./- command.

That command can be interpreted as: concatenate the - file inside the current working directory (currently working inside home directory).

For reference, you can the StackOverflow discussion here: How to open a “-” dashed filename using terminal?

Continue to the next level using the password.

ssh bandit2@bandit.labs.overthewire.org -p 2220

Level 2 → 3
The password is stored inside the spaces in this filename file located in the home directory. Consequently, I enclosed the filename in the quotation mark when using the cat command.

cat “spaces in this filename”

In Linux, quotation marks allow values containing spaces to be treated as a single argument by the command.

When I got the password, I continued my journey to level 3.

ssh bandit3@bandit.labs.overthewire.org -p 2220

Level 3 → 4
The password for level 4 is stored in a hidden file inside inhere directory.

To check the hidden file, I use the -a option for the ls command to list all the files inside the directory.

ls ./inhere -a

The ls -a command will show the hidden file name and I can directly use cat command to read the file content.

cat ./inhere/<filename>

Continue to level 4 after getting the password.

ssh bandit4@bandit.labs.overthewire.org -p 2220

Level 4 → 5
There are 10 files inside the inhere directory and only one of them is human readable.

There are two ways to solve this problem.

The first one is the most inefficient way, just cat one file at a time until we find the readable content.

The second way is to check the file type using file command and then read the one that only contains readable ASCII characters.

Use file ./inhere/* to check the file type.

As you can see, the -file07 is the one with ASCII text. Thus, we can simply use cat ./inhere/-file07 to get the password. Then, continue to the next level

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ssh bandit5@bandit.labs.overthewire.org -p 2220

Level 5 → 6
The next password is stored in a file somewhere under the inhere directory and has all of the following conditions match:

human-readable
1033 byte size
non-executable
Following those conditions, I tried to run the du -ab command combined with grep to filter the result

du ./inhere/ -ab | grep 1033

Luckily, that command returns literally 1 file and I can read the file directly to get the level 6 password.

Continue the challenge

ssh bandit6@bandit.labs.overthewire.org -p 2220

Level 6 → 7

The password is stored somewhere in the server with the following properties:

Owned by user bandit7
Owned by group bandit6
Having size 33 byte
To fulfill that requirement, I tried to find the file from the / directory using find command

find ./* -group bandit6 -user bandit7 -size -34c

Find command already provides a sophisticated option to search files for matching pattern, permissions, owner, and even size. However, the size used in the -size option will match any file with size in the range of 0 to specified -size attribute — 1

To prevent the stderr response, i.e. “permission denied”, and make the result file become obvious, redirect the stderr to an unused file in the /dev/null using additional 2>/dev/null after running the command.

In the end, this is my full command to read the password:
cat $(find ./* -group bandit6 -user bandit7 -size -34c 2>/dev/null)

After getting the password, continue to the next level.

ssh bandit7@bandit.labs.overthewire.org -p 2220

Level 7 → 8
The password is stored inside a data.txt file, next to the word millionth.

The solution is quite simple. I only used cat and grep commands to solve this.

cat data.txt | grep “millionth”

Once I got the password, I continued to the next level.

ssh bandit8@bandit.labs.overthewire.org -p 2220

Level 8 → 9
The password is stored inside the data.txt file. The password criteria is: the password is the only line of text that occurs only once.

To solve this, I combined sort, uniq, and grep command.

uniq command won't detect repeated lines if they're not adjacent. Thus, I need to sort the file content before counting the text appearance.

sort data.txt | uniq -c | grep "1 "

I used grep command to filter only 1 count text appearance result.

I continued to the next level after I got the password.

ssh bandit9@bandit.labs.overthewire.org -p 2220

Level 9 → 10
In my opinion, the difficulty of this challenge increased drastically compared to the previous ones.

The password is stored inside a data.txt file and preceded by several '=' characters. However, the data.txt file is not a fully readable ASCII character. Therefore, you need to convert it into some readable ASCII character before searching for the pattern.

I tried many commands recommended by OverTheWire, i.e. strings, base64, and xxd. However the only command that satisfies my need is the strings command.

strings command is used to print the sequences of printable ASCII characters in files. With the help of grep and regex, I could get the password.

strings data.txt | grep -Pe ‘[=]{2,}’

I used the grep command with Perl Compatible Regular Expression to find the password and took note of it and continued to the next level.

ssh bandit10@bandit.labs.overthewire.org -p 2220